

HEALTHCONNECT CLINIC - EXECUTIVE SUMMARY

Patient Attendance Analysis

Data Analytics Track

AnalystLab Africa Experience Lab

THE PROBLEM

HealthConnect Clinic experiences a 48.5% no-show rate for appointments, resulting in:

- Wasted appointment slots
- Reduced operational efficiency
- Poor patient outcomes
- Lost revenue opportunities

THE APPROACH

Analyzed 5,000 appointment records to identify:

- Key drivers of no-shows
- High-risk patient segments
- Effective interventions
- Opportunities for improvement

KEY FINDINGS (VALIDATED)

Metric	Value
No-Show Rate	48.5%
Attended Rate	46.3%
Total Appointments	5,000
Reminder Effectiveness	4.0% reduction
Missing Values	1,516

Duplicates 0

RECOMMENDATIONS

1. TARGET HIGH-RISK PATIENTS (Priority: High)
Action: Proactive outreach for patients with 2+ previous no-shows
Impact: Reduce overall no-show rate by 10-15%
2. REDUCE BOOKING LEAD TIME (Priority: High)
Action: Offer shorter wait times or same-week appointments
Impact: Reduce no-show rate by 5-10%
3. EXPAND REMINDER PROGRAMME (Priority: Medium)
Action: Send reminders via multiple channels
Impact: Additional reduction in no-shows
4. OPTIMIZE SCHEDULING (Priority: Medium)
Action: Schedule more appointments on low no-show days
Impact: Better slot utilization
5. TRANSPORT SUPPORT (Priority: Low)
Action: Partner with transport services for distant patients
Impact: Reduce no-shows for patients > 15km

BUSINESS IMPACT

- Potential reduction in no-shows
- Better appointment slot utilization
- Improved patient outcomes
- Better use of clinic resources

CROSS-TRACK COLLABORATION

Collaborated with: Data Science Track

Provided:

- Validated KPIs
- Segment analysis
- Feature recommendations

Received:

- Feature importance from model

- Model performance insights

Result:

- Better understanding of key predictors
- Validated insights for recommendations

NEXT STEPS

- Implement recommended interventions
- Monitor KPIs monthly
- Expand analysis to additional clinics
- Prepare for production deployment

Prepared By: Emmanuel Atuns

Date: 27/09/2026

Role: Data Analytics Intern, AnalystLab Africa